Answers - Sum or difference of cubes (page ??)

1. (a) $x^3 - 4^3$

   $(x-4)(x^2+4x+16)$

   (b) $x^3 + 3^3$

   $(x+3)(x^2-3x+9)$

   (c) $3(x^3-8) = 3(x^3-2^3)$

   $3(x-2)(x^2+2x+4)$

   (d) $27x^3 - 8 = (3x)^3 - 2^3$

   $(3x-2)(9x^2+6x+4)$

   (e) $8x^3 + 125 = (2x)^3 + 5^3$

   $(2x+5)(4x^2-10x+25)$

2. Write as a difference of cubes:

   $\omega^3 - 1 = 0$

   Factorise:

   $(\omega-1)(\omega^2+\omega+1) = 0$

   Since the condition stated $\omega \neq 1$, we know $\omega - 1 \neq 0$. Therefore, the other factor, $\omega^2 + \omega + 1$ must equal zero, as required.

3. LHS $= \frac{(\sin(x)+\cos(x))(\sin^2(x)-\sin(x)\cos(x)+\cos^2(x))}{\sin(x)+\cos(x)} =$

   $= \sin^2(x) - \sin(x)\cos(x) + \cos^2(x)$

   $= 1 - \sin(x)\cos(x)$

   $= 1 - \frac{1}{2}\sin(2x)$

4. $\int \frac{x^3+(2)^3}{x+2}\,dx$

   $\int \frac{(x+2)(x^2-2x+4)}{x+2}\,dx$

   $\int x^2 - 2x + 4\,dx$

   $\frac{x^3}{3} - x^2 + 4x + c$

5. Find all real solutions to the equation:

   $x^6 - 7x^3 - 8 = 0$

   Use substitution $u = x^3$

   $u^2 - 7u - 8 = 0$

   $(u - 8)(u + 1) = 0$

   Reverse substitution:

   $(x^3 - 8)(x^3 + 1) = 0$

   $(x - 2)(x^2 + 2x + 4)(x + 1)(x^2 - x + 1) = 0$

   Neither $x^2 + 2x + 4$ nor $x^2 - x + 1$ have real roots, therefore the only real solutions to the equation are $x = 2, -1$